

Jialin Zhuang



Phone: +86-15521620216 E-mail: jzhuangag@connect.ust.hk

Education

- ✧ **Sun Yat-sen University** (09/10/2020 - 28/06/2024)
 - Bachelor in School of Intelligent Systems Engineering
 - Supervisor: Professor - Shimin Gong; GPA: 3.9 / 4.0 (**range: 10/226**)
- ✧ **The Hong Kong University of Science and Technology** (02/09/2024 - present)
 - Ph.D student in Department of Electronic and Computer Engineering
 - Supervisor: Professor - Vincent Kin Nang LAU (IEEE fellow)

Research Profile & Keywords

LLM Agent Post-Training; RL Fine-Tuning; Optimizer Stability and Stepsize Design; Deep Reinforcement Learning; Stochastic Optimization; Edge Computing; Network Optimization.

Methods/Tools: PyTorch, actor-critic RL, Lyapunov drift analysis, LoRA fine-tuning, PPO/GRPO-style policy optimization, adaptive KL control, optimizer-stability diagnostics.

Selected Research Project

- Agent/RL Post-Training Optimizer Stability: Dynamic Stepsize Control (Ongoing)
- Exploring a method for LLM Agent RL post-training that treats each policy update as a risk-controlled stepsize decision under noisy trajectory rewards and KL constraints.
 - Designed a lightweight update-risk estimator using KL drift, policy-ratio deviation, advantage variance, gradient norm, and update norm, then dynamically adjusts learning rate, clip range, KL penalty, and momentum reset in PPO/GRPO-style fine-tuning.
 - Implemented the controller as an optimizer wrapper on top of AdamW/LoRA updates, focusing on optimizer stability and training dynamics rather than a full Agent application pipeline.
 - Applying the method to small open-source LLMs with reasoning/tool-use style rollouts to gain practical experience with LLM post-training workflows.

Papers and patents

1. **J. Zhuang** and V. Lau, "Uncertifiable Magnitude, Certifiable Allocation: Trust-Region Control Allocation between Improvement and Anchoring in AdamW". (Ongoing)
2. **J. Zhuang** and V. Lau, "Delay-Aware Optimizer Design for Distributed Multi-Agent Actor-Critic Learning". (Ongoing)
3. **J. Zhuang** and V. Lau, "Lyapunov-Drift Adaptive Field--Curvature Updates: Step-Size Selection and Geometry for Stochastic Saddle Fields in Robust Reinforcement Learning". (Ongoing)
4. **J. Zhuang** and V. Lau, "Momentum-Based Single-Timescale Deep Actor-Critic Reinforcement Learning: A Lyapunov

Approach for Stepsize Design and Analysis". (submitted to IEEE Trans. Signal Process.)

5. **J. Zhuang**, L. Li, Y. Long, B. Gu, C. Yi, and S. Gong, "Lagrangian-augmented learning for stochastic age of accurate semantic information minimization in mobile edge computing systems," IEEE Trans. Cogn. Commun. Netw., vol. 12, pp. 5351-5366, Jan. 2026.
6. Y. Long, **J. Zhuang**, S. Gong, B. Gu, J. Xu, and J. Deng, "Exploiting deep reinforcement learning for stochastic AoI minimization in multi-UAV-assisted wireless networks," in proc. IEEE WCNC, Dubai, United Arab Emirates, Apr. 2024, pp. 1-6.
7. S. Gong, Y. Long, **J. Zhuang**, and G. Tan, "A method and device for updating information of multi-UAVs based on DRL", Chinese Patent, CN202311518988.6, granted.
8. S. Gong, Y. Long, **J. Zhuang**, and L. Cui, "Multi-UAV-assisted data collection and transmission methods, systems, equipment, and media", Chinese Patent, CN202211207707.0, granted.
9. S. Gong, S. Zhao, S. Chen, and **J. Zhuang**. "Multi-mode control and trajectory optimization method and system for IRS carried by UAV", Chinese Patent, CN202310072734.X, granted.

Honors and awards

1. Outstanding graduate student of Sun Yat-sen University in 2024;
2. First class scholarship for Outstanding students of Sun Yat-sen University in 2023;
3. Second prize of the 10th "Datang Cup" National College Students New Generation ICT Competition in 2023;
4. Second class scholarship for Outstanding students of Sun Yat-sen University in 2022;
5. Second prize of National Undergraduate Mathematical Modeling Competition in 2022.